

System Test Cases

This document is the end-to-end test case catalog for the AMW Performance Evaluation System. Use it as the shared QA checklist for release testing, manual UAT, and automated coverage planning.

Test Strategy

Layer	Command	Purpose
Backend unit	<code>npm run test -w backend</code>	Pure business logic, middleware helpers, OpenAPI generation, scoring, password utilities
Backend integration	<code>npm run test:integration:setup && npm run test:integration -w backend</code>	API behavior against migrated and seeded PostgreSQL
Frontend unit	<code>npm run test -w frontend</code>	Pure UI logic, scoring helpers, constants, access policy helpers
Frontend E2E	<code>npm run test:e2e -w frontend</code>	Browser workflows with mocked API responses
Full QA robot	<code>npm run qa</code>	Running application checks across auth, navigation, dashboard, forms, reports, RBAC, responsive viewports
Release gate	<code>npm run verify</code>	Lint, unit, E2E, build, and audit gate

Test Data

Account	Role	Main Use
<code>developer@amw-ems.com</code>	Developer	Super-role, admin-level override, release validation
<code>admin@amw-ems.com</code>	Admin	User management, reports, calibration, destructive admin actions
<code>manager.eng@amw-ems.com</code>	Manager	Create evaluations, review team data, reports
<code>supervisor1@amw-ems.com</code>	Supervisor	Create and complete evaluations
<code>officer1@amw-ems.com</code>	Employee	Self profile, own evaluations, restricted access checks

Passwords must come from the local seed configuration and must not be copied into public documentation outside this repository.

Entry And Exit Criteria

Entry:

- Test database is migrated and seeded.
- Frontend and backend build locally.
- Environment variables match `.env.example`, `.env.test`, or the target release profile.
- Browser dependencies for Playwright are installed.

Exit:

- All critical and high severity test cases pass.
- No unresolved P0/P1 bugs remain.
- All failed medium or low severity cases have an owner and release decision.
- `npm run verify` passes or an approved exception is recorded.

Severity

Severity	Definition
Critical	Blocks login, exposes sensitive data, breaks role isolation, corrupts evaluation data, or prevents release
High	Breaks a main workflow for a major role or causes incorrect scores/reports
Medium	Breaks a secondary workflow, validation, usability, or observability expectation
Low	Cosmetic or minor copy/formatting issue with no workflow impact

Authentication And Session

ID	Scenario	Steps	Expected Result	Severity	Automation
AUTH-001	Login with valid company email	Open login, enter admin credentials, submit	User lands on dashboard with valid session	Critical	Backend integration, QA robot
AUTH-002	Login with employee number	Enter seeded employee number and password	Login succeeds and user profile loads	High	Backend integration
AUTH-003	Invalid password	Enter valid identifier with wrong password	API returns 401 and UI shows safe error	Critical	Backend integration, QA robot
AUTH-004	Unknown identifier	Enter unknown email or employee number	API returns 401 without account discovery details	High	Backend integration
AUTH-005	Empty login body	Submit empty request	API returns 400 validation error	Medium	Backend integration
AUTH-006	Malformed token	Call protected route with fake token	API returns 401	Critical	Backend integration
AUTH-007	Expired token	Call protected route with expired JWT	API returns 401 and frontend signs user out	Critical	Backend integration
AUTH-008	Session persistence	Refresh browser after login	User remains signed in until expiry	High	E2E
AUTH-009	Session expiry warning	Keep session near expiry	Warning appears and logout path works	Medium	QA robot
AUTH-010	Logout	Click logout	Session storage is cleared and protected pages redirect to login	High	QA robot
AUTH-011	Forgot password mismatch	Submit employee number and wrong birth date	Reset is rejected with no token or sensitive details	High	Backend integration
AUTH-012	Forgot password match	Submit matching employee number and birth date with strong new password	Password changes, must-change flag clears, login succeeds	Critical	Backend integration
AUTH-013	Buddhist year birthday input	Reset imported user with Buddhist year date	Reset succeeds when date maps to source birthday	High	Backend integration
AUTH-014	Weak password	Register or reset with weak password	API returns validation error	High	Backend unit/integration
AUTH-015	Non-company email registration	Attempt registration with external domain	API rejects the request	High	Backend integration

Role And Access Control

ID	Scenario	Steps	Expected Result	Severity	Automation
RBAC-001	Developer can access privileged surfaces	Sign in as developer and open users, reports, evaluations	All privileged routes load	Critical	Integration, QA robot
RBAC-002	Admin can manage users	Sign in as admin and open user management	User table and admin actions are visible	Critical	Integration, QA robot
RBAC-003	Manager cannot access admin user management	Sign in as manager and request <code>/api/users</code>	API returns 403	Critical	Integration
RBAC-004	Employee cannot access admin user management	Sign in as employee and request <code>/api/users</code>	API returns 403	Critical	Integration
RBAC-005	Employee cannot create evaluation	POST evaluation as employee	API returns 403	Critical	Integration
RBAC-006	Supervisor can create evaluation	POST evaluation as supervisor for existing employee	Evaluation is created as DRAFT	High	Integration
RBAC-007	Admin cannot edit evaluator-owned form sections	Save goals or answers as admin	API returns 403 where <code>blockAdmin</code> applies	High	Integration
RBAC-008	Non-participant cannot read evaluation	Supervisor opens unrelated evaluation	API returns 403	Critical	Integration
RBAC-009	Participant can read evaluation	Evaluator or evaluatee opens own evaluation	API returns 200	Critical	Integration
RBAC-010	Admin/developer can delete evaluation	Delete evaluation as admin or developer	API returns 204 and record is removed	High	Integration, E2E
RBAC-011	Manager/employee cannot delete evaluation	Delete evaluation as non-admin	API returns 403	Critical	Integration
RBAC-012	Employee cannot change HR-owned profile fields	PATCH <code>/api/auth/me</code> with position or role	API returns 400 or 403	Critical	Integration
RBAC-013	Report access follows role policy	Request summary and audit reports across roles	Only authorized roles receive data	High	Integration, QA robot

Evaluation Workflow

ID	Scenario	Steps	Expected Result	Severity	Automation
EVAL-001	Evaluation list loads	Open evaluations page as admin	List shows seeded and created evaluations	Critical	Integration, E2E
EVAL-002	Employee scoped list	Open evaluations as employee	Only own evaluator/evaluatee records appear	Critical	Integration
EVAL-003	Create evaluation for existing employee	Select cycle, evaluator, evaluatee, form type	New DRAFT evaluation appears	Critical	Integration, E2E
EVAL-004	Create evaluation with new evaluatee as admin	Submit new evaluatee data	User and evaluation are created	High	Integration
EVAL-005	Reject duplicate or invalid create	Submit duplicate or missing evaluatee payload	API returns 400 or 409	High	Integration
EVAL-006	Save draft answers	Fill rating/text answers and save	Status moves from DRAFT to IN_PROGRESS	Critical	Integration
EVAL-007	Save goals	Add goal entries with weights and scores	Entries persist and reload correctly	Critical	Integration
EVAL-008	Save competency scores	Enter all required competency scores	Scores persist and validation state updates	Critical	Integration, QA robot
EVAL-009	Save attendance	Enter leave, late, and disciplinary values	Attendance score is calculated and saved	Critical	Integration
EVAL-010	Save training section	Add training requirements or completed training	Data persists and contributes to readiness where applicable	Medium	QA robot/manual
EVAL-011	Save comments	Add strengths, improvements, required skills	Comments persist and export includes them	Medium	QA robot/manual
EVAL-012	Save salary summary	Manager/developer enters salary and bonus fields	Sensitive salary data persists only for allowed roles	Critical	Integration/manual
EVAL-013	Submit complete evaluation	Complete required sections and submit	Status becomes SUBMITTED or PENDING_REVIEW	Critical	Integration
EVAL-014	Block incomplete submit	Leave required sections incomplete and submit	UI/API blocks submit with clear missing section state	High	QA robot/manual
EVAL-015	Two-stage reviewer submit	Submit evaluation with reviewer, then reviewer approves	Status moves PENDING_REVIEW to SUBMITTED	Critical	Integration
EVAL-016	Non-reviewer cannot review	Call review endpoint as unrelated user	API returns 400 or 403	Critical	Integration
EVAL-017	Acknowledgement signing	Manager saves employee/evaluator/director sign-off dates	Dates persist and re-saving is idempotent	High	Integration
EVAL-018	Delete evaluation	Delete as admin from UI	Row disappears and API returns 204	High	Integration, E2E
EVAL-019	Unknown evaluation	Request missing evaluation id	API returns 404	Medium	Integration

ID	Scenario	Steps	Expected Result	Severity	Automation
EVAL-020	Form type selection by position	Create evaluations for officer, supervisor, manager, director	Correct form definition and competencies load	High	Manual/QA robot

Scoring And Calibration

ID	Scenario	Steps	Expected Result	Severity	Automation
SCORE-001	Empty score input	Calculate total from no rating answers	Returns null	High	Backend unit
SCORE-002	Equal weighted average	Score two equal weighted ratings	Returns simple average	Critical	Backend unit
SCORE-003	Weighted score	Score ratings with unequal weights	Returns weighted average	Critical	Backend unit
SCORE-004	Null and text answers	Mix null/text answers with rating answers	Non-rating and null scores are ignored	High	Backend unit
SCORE-005	Goal weight validation	Enter goal weights that do not total required amount	UI/API blocks submit or marks not ready	High	QA robot/manual
SCORE-006	Configurable scoring weights	Change evaluation weights	Total score reflects goal, competency, attendance, and training weights	Critical	Unit/manual
SCORE-007	GPA final grade	Submit evaluation with final total score	GPA-style grade equals configured final grade mapping	Critical	Unit/manual
SCORE-008	Calibration page loads	Admin opens calibration workspace	Cycle scores and final grade controls load	High	QA robot/manual
SCORE-009	Calibrate final grade	Admin changes final grade	Grade saves and appears in reports/export	High	Manual
SCORE-010	Non-admin calibration blocked	Manager or employee attempts grade calibration	API/UI denies action	Critical	Integration/manual

Templates

ID	Scenario	Steps	Expected Result	Severity	Automation
TPL-001	Template list	Open templates page	Existing templates render	High	Integration, E2E
TPL-002	Create template	Click New Template and submit required fields	Template is created and detail page opens	High	E2E
TPL-003	Edit template metadata	Change name, description, type	Changes persist	Medium	Manual
TPL-004	Add section	Add section to template	Section appears in order	Medium	Manual
TPL-005	Add question	Add rating/text question	Question appears with correct type and weight	Medium	Manual
TPL-006	Delete template	Delete unused template as admin	Template is removed	Medium	E2E
TPL-007	Delete protected template	Delete template used by active cycle	API rejects or safely handles dependency	High	Manual
TPL-008	Role access to template builder	Open builder across roles	Only allowed supervisory/admin roles can mutate	High	QA robot/manual

Cycles

ID	Scenario	Steps	Expected Result	Severity	Automation
CYCLE-001	Cycle list	Open cycles page across roles	Active cycles load	High	Integration, E2E
CYCLE-002	Create cycle	Select template and valid dates	Cycle is created	High	E2E
CYCLE-003	Invalid date range	End date before start date	Validation blocks save	High	Manual
CYCLE-004	Status change	Admin changes cycle status	Status persists and affects available actions	High	Manual
CYCLE-005	Delete cycle	Delete fixture cycle with no blocking data	API returns 204	Medium	E2E/manual
CYCLE-006	Delete cycle with evaluations	Delete cycle that owns evaluations	API rejects or cascades only by approved policy	High	Manual

Reports And Export

ID	Scenario	Steps	Expected Result	Severity	Automation
REP-001	Summary report	Open reports page as admin	Cycle summary and department breakdown load	Critical	Integration, E2E
REP-002	Dashboard report stats	Open dashboard	Total, completed, pending, average score, users, active cycles show	High	E2E
REP-003	CSV export	Export evaluation CSV	Download filename and CSV content are correct	High	E2E
REP-004	PDF export English	Export evaluation PDF in English	PDF downloads with correct filename and content	High	E2E/manual
REP-005	PDF export French	Export evaluation PDF in French	PDF downloads with French filename/content	High	E2E/manual
REP-006	Thai PDF font	Export Thai content	Thai glyphs render without tofu or garbling	High	Manual
REP-007	Salary export boundary	Export as employee	Salary fields are omitted or denied	Critical	Manual
REP-008	Audit events report	Open audit report as admin	Login/export/mutation events appear with request id	Critical	E2E
REP-009	Export audit trail	Export a report and inspect audit records	Export event is persisted with actor and target	Critical	Integration/manual

User And Data Management

ID	Scenario	Steps	Expected Result	Severity	Automation
USER-001	User list	Admin opens users page	User table loads without password hashes	Critical	Integration, QA robot
USER-002	Create user	Admin creates user with valid company email	User is created with temporary password policy	Critical	Manual
USER-003	Duplicate user	Create existing email or employee number	API returns conflict	High	Manual
USER-004	Update user profile	Admin edits department, position, role	Changes persist and access policy updates	High	Manual
USER-005	Disable/delete user	Admin disables or deletes eligible user	User can no longer access as intended	High	Manual
USER-006	Self profile update	Employee updates own editable fields	Allowed fields persist	Medium	Manual
USER-007	Employee import	Import employee CSV/source data	Valid rows create/update users and invalid rows are reported	High	QA robot/manual
USER-008	Temporary password reissue	Run reissue script for target users	New temporary passwords are generated and must-change flag is set	High	Manual

Localization And UI

ID	Scenario	Steps	Expected Result	Severity	Automation
UI-001	Thai locale	Switch to Thai and navigate major pages	Labels render in Thai and layout remains stable	Medium	QA robot/manual
UI-002	English locale	Switch to English	Labels render in English	Medium	E2E
UI-003	French locale	Switch to French	Labels render in French and exports offer French PDF	Medium	E2E/manual
UI-004	Theme switch	Toggle light/dark mode	Theme persists after refresh	Low	QA robot/manual
UI-005	Command palette	Press Ctrl+K and search Swagger	Admin command appears and can be selected	Medium	E2E
UI-006	Toast feedback	Complete create/delete/save actions	Success/error toast appears and disappears correctly	Low	Manual
UI-007	Error boundary	Trigger recoverable frontend error	Error boundary displays safe fallback	Medium	Manual
UI-008	Empty states	View pages with no data	Empty state is clear and action buttons are appropriate	Low	Manual

Responsive And Accessibility

ID	Scenario	Steps	Expected Result	Severity	Automation
A11Y-001	Dashboard no critical violations	Run axe on dashboard	No critical accessibility violations	High	E2E
A11Y-002	Reports no critical violations	Run axe on reports	No critical accessibility violations	High	E2E
A11Y-003	Keyboard navigation	Navigate login, command palette, forms, modals with keyboard	Focus order is logical and visible	High	Manual
A11Y-004	Modal focus trap	Open create/delete dialogs	Focus stays inside modal and Escape/close works	Medium	Manual
A11Y-005	Desktop viewport	Run key flows at 1440x900	No horizontal overflow or clipped controls	High	E2E, QA robot
A11Y-006	Tablet viewport	Run key flows at 768x1024	Navigation, tables, dialogs remain usable	High	QA robot
A11Y-007	Mobile viewport	Run key flows at 390x844	No overlapping controls or unusable modals	High	QA robot
A11Y-008	Long text	Enter long names/departments/comments	Text wraps or truncates without layout breakage	Medium	Manual

API, Observability, And Operations

ID	Scenario	Steps	Expected Result	Severity	Automation
OPS-001	Liveness	GET <code>/health</code>	200 with version, env, release, request id	Critical	Integration
OPS-002	Readiness healthy	GET <code>/api/ready</code> with DB up	200 with database <code>ok</code>	Critical	Integration/manual
OPS-003	Readiness degraded	Stop DB and request readiness	503 with database <code>degraded</code>	Critical	Manual
OPS-004	Metrics public/no token mode	GET <code>/metrics</code> in local config	Prometheus text response returns	Medium	Manual
OPS-005	Metrics token guard	Set <code>METRICS_TOKEN</code> and call without token	401; valid Bearer token succeeds	High	Manual
OPS-006	Request id propagation	Send request with/without request id	Response and logs include request id	High	Unit/integration
OPS-007	No-store API headers	Call protected API route	Cache-Control includes no-store	High	Integration
OPS-008	Helmet headers	Call API and inspect headers	Security headers are present and X-Powered-By absent	High	Integration
OPS-009	Swagger UI	Open <code>/api/docs/</code>	Swagger UI renders	Medium	Manual
OPS-010	OpenAPI JSON	GET <code>/api/openapi.json</code>	Valid OpenAPI document returns	High	Backend unit/manual
OPS-011	Docker compose demo	Start demo compose	Frontend, backend, migration, and database become healthy	High	Manual
OPS-012	Production compose	Start production compose with <code>.env.prod</code>	Migration runs once and app serves through nginx	Critical	Manual

Security And Abuse

ID	Scenario	Steps	Expected Result	Severity	Automation
SEC-001	IDOR read prevention	Employee requests unrelated evaluation id	API returns 403	Critical	Integration
SEC-002	IDOR delete prevention	Employee deletes unrelated evaluation	API returns 403	Critical	Integration
SEC-003	JWT signature tamper	Alter valid token signature	API returns 401	Critical	Integration
SEC-004	JWT role elevation	Forge token claiming admin role	API returns 401	Critical	Integration
SEC-005	Missing Bearer prefix	Send token without <code>Bearer</code>	API returns 401	High	Integration
SEC-006	SQL/XSS login payloads	Submit injection strings to login	API returns 4xx and never 500	High	Integration
SEC-007	Long input	Submit oversized identifier/name values	API rejects safely or returns non-500	High	Integration
SEC-008	Password hash exposure	Inspect login, <code>/me</code> , and user list responses	No password hash or bcrypt marker appears	Critical	Integration
SEC-009	Auth rate limit	Send repeated failed login requests	API eventually returns 429	High	Integration
SEC-010	CORS allowlist	Request from unapproved origin	Browser/API denies credentialed access	High	Manual
SEC-011	Audit log for mutations	Create/update/delete evaluation	Audit event records actor, path, target, status	Critical	Integration/manual
SEC-012	Public registration guard	Disable public registration and attempt register	API rejects when production guard is active	High	Manual

Performance And Reliability

ID	Scenario	Steps	Expected Result	Severity	Automation
PERF-001	Dashboard load	Open dashboard with seeded dataset	First meaningful content appears within accepted target	Medium	QA robot/manual
PERF-002	Evaluation list load	Open evaluations with many records	Page remains responsive and paginated/filterable if applicable	Medium	QA robot/manual
PERF-003	Report generation	Generate summary and export report	API responds within accepted target and does not time out	High	Manual
PERF-004	Concurrent logins	Run multiple login attempts with valid users	No 5xx and rate limiting does not block normal users	High	Manual
PERF-005	Concurrent section saves	Save goals/competency/attendance rapidly	Last accepted save wins without corrupting related sections	High	Manual
PERF-006	Restart recovery	Restart backend while frontend is open	User sees recoverable error and can continue after API returns	Medium	Manual
PERF-007	Backup restore drill	Restore DB backup to test environment	Application starts and critical data is present	Critical	Manual

Regression Suites

Run these groups before merging high-risk changes:

Change Area	Minimum Regression
Auth, session, password reset	AUTH-*, SEC-003 to SEC-009 , backend auth integration
Roles or access policy	RBAC-*, SEC-001 , SEC-002 , role matrix integration
Evaluation form or scoring	EVAL-*, SCORE-*, frontend E2E export flow
Reports or exports	REP-*, SEC-011 , Thai/French PDF manual checks
Templates or cycles	TPL-*, CYCLE-*, E2E create/delete actions
Layout, i18n, navigation	UI-*, A11Y-*, Playwright E2E, QA robot mobile/tablet suites
Deployment or environment	OPS-*, production readiness checklist

Known Manual Coverage

These cases are intentionally manual today because they require real rendered PDF inspection, production-like infrastructure, large datasets, or business sign-off:

- Thai/French PDF visual correctness.
- Production Docker and nginx deployment validation.
- Database backup restore drill.
- Calibration approval policy with HR sign-off.
- Large import and report performance tests.
- CORS validation against real deployed origins.